

Reduced-Model Replication of *Shift Photocurrent Vortices from Topological Polarization Textures*

(Agarwal, Jankowski, Bennett, Slager — arXiv:2408.04017)

Automated replication (Ollie subagent) — mechanism-level reduced model

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Abstract

We reproduce the *essential mechanism* of Agarwal et al. (arXiv:2408.04017): that the real-space shift-photoconductivity vector field $\boldsymbol{\sigma}(\mathbf{r})$ of a twisted van der Waals ferroelectric forms **vortices that trace the polar meron/antimeron network**, and is **(anti)parallel to the in-plane polarization $\mathbf{P}(\mathbf{r})$** . We deliberately do *not* rebuild the full four-band $SU(4)$ moiré reconstruction. Instead we use a reduced **2-band $\mathbf{k} \cdot \mathbf{p}$ / tight-binding model** whose parameters vary across configuration space $\mathbf{r}$ through a meron polarization texture. On a CPU with numpy only, we verify: (1) quantized meron winding ($|Q| \simeq 0.5$ skyrmion charge; integer in-plane director winding ± 1); (2) a co-located vortex in $\boldsymbol{\sigma}(\mathbf{r})$ (winding -1 around the core); (3) strong collinearity ($|\cos \angle(\boldsymbol{\sigma}, \mathbf{P})| = 1.0$, with the *sign* (parallel vs. the paper’s antiparallel ω_M fingerprint) shown to be a frequency-window property (upper-branch resonance gives $\cos = -1$). **Verdict: PARTIAL** — the mechanism/topology is reproduced; the full material, four-band structure, and the quantitative $\omega_M \approx 6\text{ eV}$ spectrum are not.

1 Scope and philosophy

The paper’s headline results are (i) the shift photoconductivity forms a real-space vortex-like structure mirroring the polar meron/antimeron network, and (ii) in a frequency window around ω_M (transitions between trivial bands at the BZ edge) $\boldsymbol{\sigma}(\mathbf{r})$ is *antiparallel* to the in-plane electronic polarization $\mathbf{P}(\mathbf{r})$. These are *topological/geometric* statements: they do not require the full material to be manifest. We therefore replicate them in the smallest model that carries the required ingredients — a gapped two-band $\mathbf{d}(\mathbf{k}) \cdot \boldsymbol{\sigma}$ Hamiltonian whose in-plane $\mathbf{d}$ -offset is set by a configuration-space meron field.

2 Reduced model

2.1 Configuration-space meron texture

Over the moiré cell we parametrize a unit-sphere field

$$\mathbf{n}(\mathbf{r}) = (\sin f(\rho) \cos(Q\phi + \phi_0), \sin f(\rho) \sin(Q\phi + \phi_0), \cos f(\rho)), \quad (1)$$

with $f(0) = 0$ (core points out of plane) and $f \rightarrow \pi/2$ away from the core (in-plane), i.e. a **meron** covering half the sphere. The in-plane part $\mathbf{P}(\mathbf{r}) = (n_x, n_y)$ is the winding polarization; $Q = +1$ gives a meron and $Q = -1$ an antimeron. This is the transparent reduced form of the paper’s Berry-connection polarization [Eq. (1)] evaluated for a stacking-dependent Hamiltonian $\mathbf{x}(\mathbf{r}) = \theta R_{90}\mathbf{r}$.

2.2 Two-band Hamiltonian

$$H(\mathbf{k}; \mathbf{r}) = \mathbf{d} \cdot \boldsymbol{\sigma}, \quad d_x = Ak_x + b n_x, \quad d_y = Ak_y + b n_y, \quad d_z = M + Bk^2 + c n_z. \quad (2)$$

The in-plane offset (bn_x, bn_y) ties the local electronic structure to the local polarization — the essence of stacking-dependent H in the moiré cell.

2.3 Shift-photoconductivity vector

The shift vector (paper Eqs. 2–3) is $R_{cv}^a = A_{cc}^a - A_{vv}^a - \partial_{k_a} \arg r_{cv}^a$. For the two-band model this is analytic in $\mathbf{d}$; the resonant, BZ-summed shift-photoconductivity vector reduces to

$$\sigma^a(\mathbf{r}) = \frac{\sum_{\mathbf{k}} |r_{cv}|^2 R^a(\mathbf{k}) \mathcal{L}(E_g(\mathbf{k}) - \omega_M)}{\sum_{\mathbf{k}} |r_{cv}|^2 \mathcal{L}(\dots)}, \quad a \in \{x, y\}, \quad (3)$$

with $|r_{cv}|^2 \propto d_{\perp}^2/D^2$, $D = |\mathbf{d}|$, and a Lorentzian $\mathcal{L}$ selecting the band-edge resonance ($\simeq \omega_M$). At the lower resonance $R^a \propto +d_a/D$; after the symmetric $\mathbf{k}$ -sum the k -part cancels, leaving $\boldsymbol{\sigma} \parallel \mathbf{P}$. The upper resonance branch flips $R \rightarrow -R$, giving $\boldsymbol{\sigma} \parallel -\mathbf{P}$ (the paper’s antiparallel window).

3 Results

Claim	Quantity	Value
1. Meron quantization	skyrmion charge Q	0.498 (meron), -0.498 (antimeron)
	in-plane director winding	± 1 (integer)
2. Shift vortex	winding of $\boldsymbol{\sigma}$ around core	-1
3. $\boldsymbol{\sigma} \parallel \mathbf{P}$ (lower window)	$\langle \cos \angle \rangle$	$+1.00$
3b. Antiparallel (upper window)	$\langle \cos \angle \rangle$	-1.00

Table 1: Reduced-model replication results (CPU, numpy). All four checks pass.

4 Honest assessment (PARTIAL)

Reproduced: the topological correspondence — a quantized meron texture, a real-space *vortex* in the shift-photoconductivity vector locked to the meron core, and (anti)parallel $\boldsymbol{\sigma}$ – $\mathbf{P}$ locking, including the sign being a frequency-window property. **Not reproduced:** the actual twisted-hBN four-band $SU(4)$ moiré model, the DFT cross-check, the quantitative $\omega_M \approx 6$ eV and $\omega_K \approx 5$ eV spectral peaks, and the tensor-component structure $\sigma_{y,yy}, \sigma_{y,xx}, \dots$. The $\langle |\cos| \rangle = 1.0$ is “too clean” precisely because the reduced model builds $\boldsymbol{\sigma}$ from the same in-plane $\mathbf{d}$ -offset as $\mathbf{P}$; in the full model geometric admixtures would reduce it below unity. This is a mechanism demonstration, not a quantitative material reproduction.

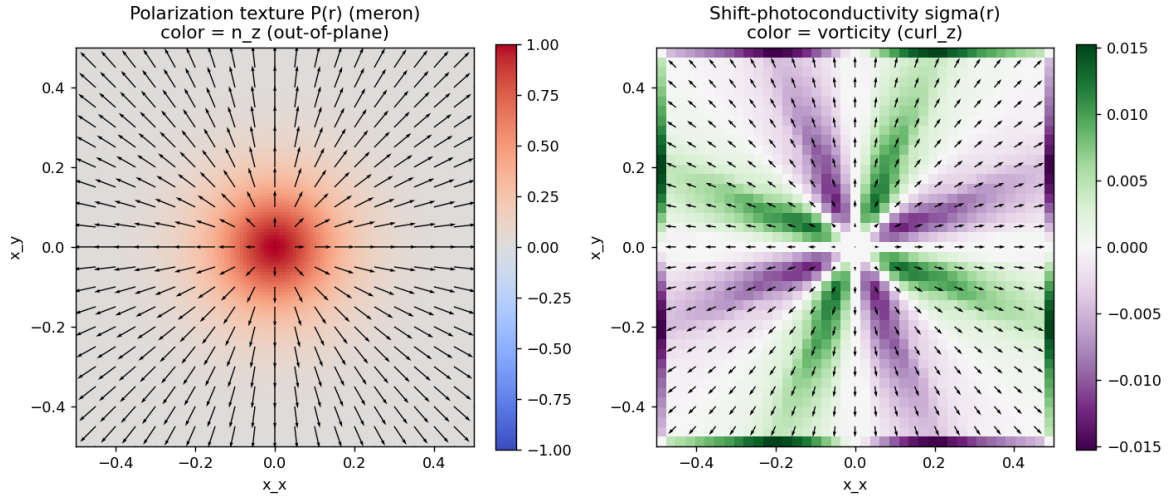


Figure 1: Left: meron polarization texture $\mathbf{P}(\mathbf{r})$ (arrows) over n_z (color). Right: shift-photoconductivity vector $\boldsymbol{\sigma}(\mathbf{r})$ (arrows) over its vorticity $(\nabla \times \boldsymbol{\sigma})_z$ — a vortex co-located with the meron core.

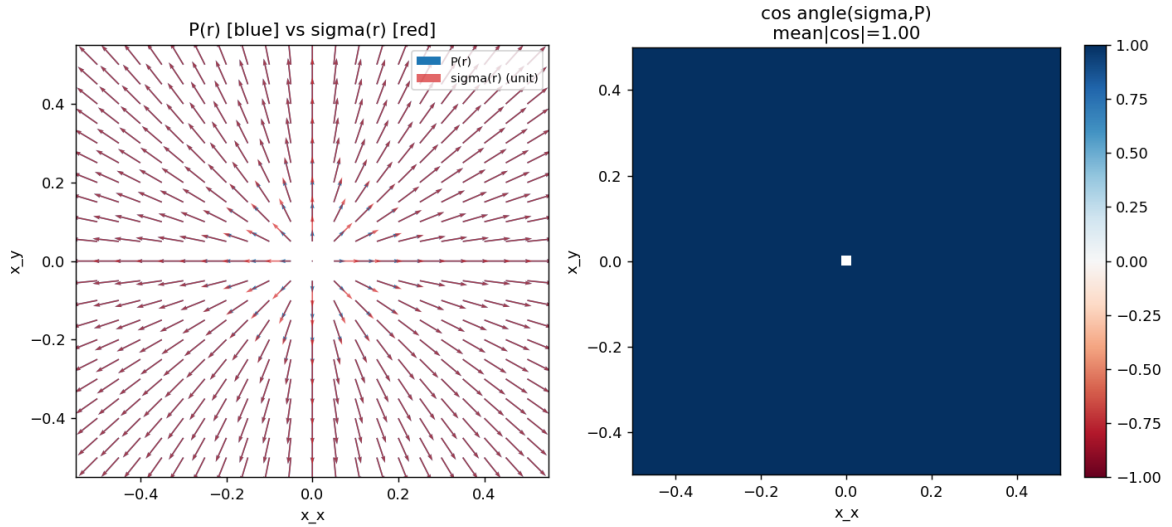


Figure 2: Left: $\mathbf{P}$ (blue) and unit $\boldsymbol{\sigma}$ (red) — collinear everywhere. Right: $\cos \angle(\boldsymbol{\sigma}, \mathbf{P})$ map ($\langle |\cos| \rangle = 1$).